

B. Architecture of the MSDAP

Based on the previous section, the proposed architecture for the MSDAP is shown in figure 4.4. Only one channel is shown for clarity. It consists of the following functional blocks:

- S2P (Serial-to-Parallel Interface).- This converts the serial input into 16-bit words that can be loaded into one of the memory units. The following are the I/O pins:
 - DCLK (input).- Data clock.
 - CLR (input).- Clear signal.
 - EN (input).- Enable.
 - DATA_IN (input).- Serial input data.
 - PDATA_IN (output).- Parallel data (16-bit word).
- Data Memory.- This memory unit stores the input audio data. It stores up to 256 16-bit words, since the maximum order of the FIR filter is 255. It also contains an all-zero detector for the last data stored. The following are the I/O pins:
 - CLEAR (input).- Clear signal coming from the main controller.
 - LOAD (input).- Load signal connected to frame.
 - PDATA_IN (input).- 16-bit input data.
 - WRITE_ADDRESS (input).- 8-bit write address, controlled by the Main Controller.
 - WRITE_EN (input).- Write Enable.
 - READ_ADDRESS (input).- 8-bit read address controlled by the ALU Controller.
 - DATAOUT (output).- 16-bit output data stored in location READ_ADDRESS.
- R_j 's Memory.- This unit stores R_j , which specifies the number of operands in the computation of u_j . This unit can store up to 16 8-bit words. The following are the I/O pins:
 - CLEAR (input).- Clear signal coming from the main controller.
 - LOAD (input).- Load signal connected to frame.
 - PDATA_IN (input).- 8-bit input data (connected to the 8LSBs from the output of the S2P).
 - WRITE_ADDRESS (input).- 4-bit write address, controlled by the Main Controller.
 - WRITE_EN (input).- Write Enable.
 - READ_ADDRESS (input).- 4-bit read address controlled by the ALU Controller.
 - DATAOUT (output).- 8-bit output data stored in location READ_ADDRESS.
- Coefficient Memory.- This memory unit stores the coefficients $a(n)$ that contain the index of the data samples required for each u_j . The MSB is the sign bit (0 means addition, 1 means subtraction) for the calculation of u_j . This unit can store up to 512 9-bit words, since the sum of R_j s is 512. The following are the I/O pins:
 - CLEAR (input).- Clear signal coming from the main controller.
 - LOAD (input).- Load signal connected to frame.
 - PDATA_IN (input).- 9-bit input data (connected to the 9LSBs from the output of the S2P).

- WRITE_ADDRESS (input).- 5-bit write address, controlled by the Main Controller.
 - WRITE_EN (input).- Write Enable.
 - READ_ADDRESS (input).- 5-bit read address controlled by the ALU Controller.
 - DATAOUT (output).- 9-bit output data stored in location READ_ADDRESS.
- Main Controller.- This is the main finite-state machine (FSM) controlling the different modes of operation of the MSDAP chip. The FSM flowchart was shown in figure 3.4.
 - START (input).- Start signal coming into the chip.
 - RESET_N (input).- Reset_n signal coming into the chip (active low).
 - SCLK (input).- System clock.
 - DCLK (input).- Data clock.
 - FRAME (input).- Frame signal coming into the chip.
 - CLEAR(output).- Clears the memory units, and reset the ALU.
 - WRITE_EN_RJ (output).- Write Enable for Rj Memory.
 - WRITE_EN_COEFF (output).- Write Enable for Coefficient Memory.
 - WRITE_EN_DATA (output).- Write Enable for Data Memory.
 - WRITE_ADDRESS_RJ (output).- 4-bit write address for Rj Memory.
 - WRITE_ADDRESS_COEFF (output).- 9-bit write address for Coefficient Memory.
 - WRITE_ADDRESS_DATA (output).- 8-bit write address for Data Memory.
 - EN_FIR (output).- Enable the ALU.
- Arithmetic-Logic Controller (ALU).- This unit computes the FIR filter output based on the algorithm proposed in [1]. It itself contains the following sub-blocks:
 - Sign Extender.- Extends the 16-bit input data words to 24-bits using the sign bit.
 - DATAIN (input).- 16-bit data coming from the Data Memory.
 - DATAOUT (output).- 24-bit data.
 - Adder.- It is a 24-bit adder that performs the accumulation operation required by the FIR filter algorithm.
 - X1 (input).- 24-bit word input connected to the output of the sign extender.
 - X2 (input).- 24-bit word input connected to the 24MSBs of the output from the shift register.
 - ADD/SUBTRACT (input).- Sign-bit coming from the MSB of the coefficients stored in Coefficient memory. $X2+X1$ when ADD/SUBTRACT is low. $X2-X1$ when ADD/SUBTRACT is high.
 - Y (output).- 24-bit output of the adder.
 - 1-Bit Shift-Right.- It is a shift register that stores a 40-bit input and shifts it to the right by one bit per SCLK cycle when SHIFT is high. It loads the 24-bit output from the accumulator into its 24MSBs. The 16LSBs are not affected by the load operation. Only shift-right operations and CLEAR can alter the content of the 16LSBs.
 - CLEAR (input).- Clear signal.
 - SCLK (input).- System clock.
 - LOAD (input).- Load signal coming from the ALU controller.

- SHIFT (input).- Shift signal coming from the ALU controller. Shift 1-bit right when SHIFT is high.
- DATAIN (input).- 24-bit word coming from the output of the adder.
- DATAOUT (output).- 40-bit word output.
- ALU Controller.- This is the FSM that controls the operation of the ALU.
 - COEFF(input).- 8 LSBs coming from the Coefficient Memory.
 - R_j 's(input).- 8-bit word coming from the R_j Memory.
 - EN_FIR (input).- Enable the ALU.
 - CLEAR (input).- Clear the ALU contents and reset the ALU Controller.
 - READ ADDRESS(output).- 8-bit read address going to Data Memory.
 - READ ADDRESS(output).- 4-bit read address going to R_j 's Memory.
 - READ ADDRESS(output).- 5-bit read address going Coeff Memory.
 - LOAD (output).- Load the computed data in P2S
 - CLEAR_REG (output).- Clear contents of the shift register.
 - LOAD (output).- Load the output of the adder into the shift register.
 - SHIFT (output).- Indicate the shift register to shift right.
- P2S (Parallel-to-Serial Interface).- This converts the output of the FIR filter from a 40-bit word into a serial output. It uses the system clock SCLK for the transmission.
 - SCLK (input).- System clock.
 - CLR (input).- Clear signal.
 - LOAD (input).- Load the 40-bit parallel word to be transmitted.
 - EN (input).- Enable.
 - PDATAIN (input).- Parallel 40-bit input data.
 - DATAOUT (output).- Serial output data.

Notice that for clarity, not all the I/O pins for each functional block have been included in figure 4.4. Also, the main controller is shared between the two identical datapaths, left and right.

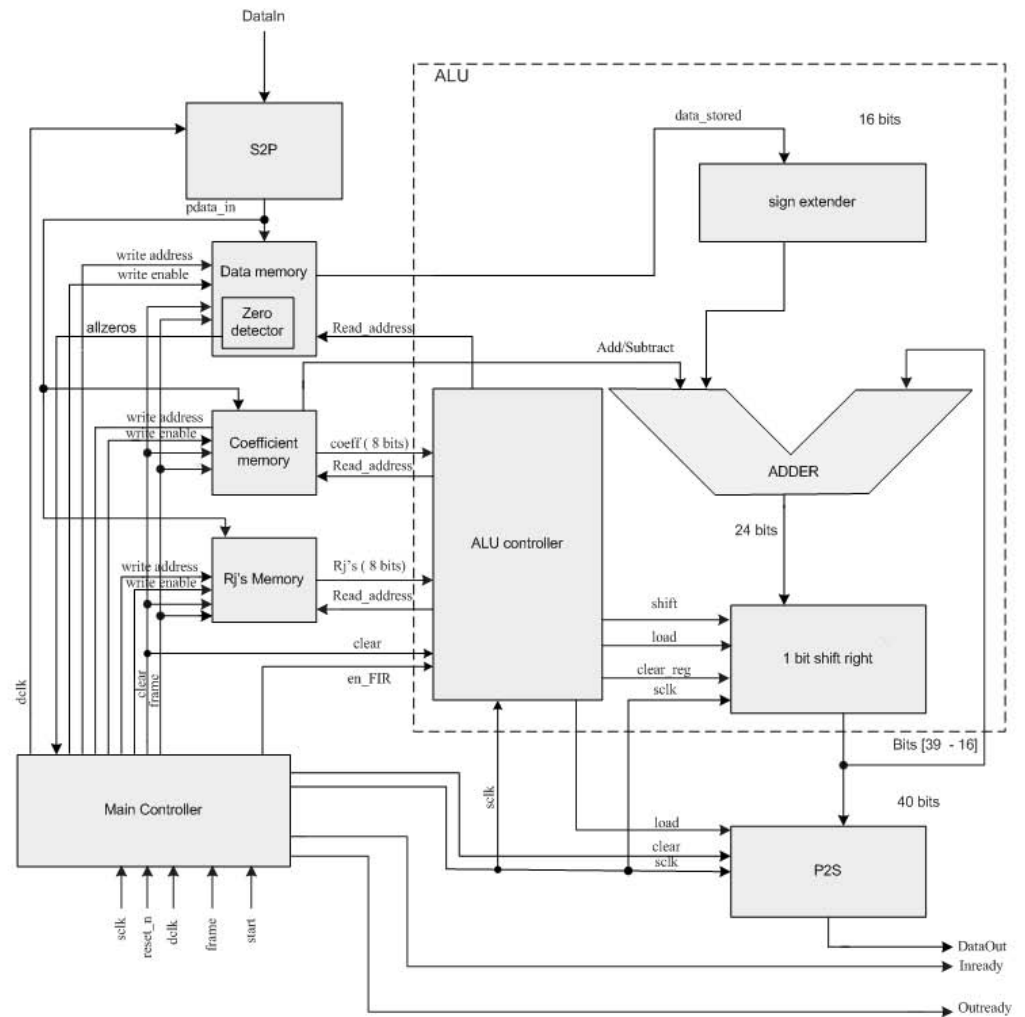


Figure 4.4. Proposed architecture for the MSDAP.

C. Time Budget

Since the frequency of the data clock is 768kHz, and each frame has 16 DCLKs, we know that each frame lasts $T_{FRAME} = \frac{16}{768k\text{Hz}} = 20.8\bar{3}\mu\text{s}$. The system clock frequency is 26.88 MHz,